

The Micro-Break Effect: Reducing Digital Eye Strain Through Short, Scheduled Device Pauses

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Abstract

Digital eye strain (DES) affects up to 65% of regular smartphone users. This controlled study evaluated whether a mandatory 20-second pause every 20 minutes (the 20-20-20 rule) reduces self-reported DES metrics over an 8-hour period. Participants using the pause intervention reported a 34% reduction in visual fatigue compared to the control group. These data validate micro-breaks as a highly effective, low-friction intervention for daily device interaction.

1. Introduction

Prolonged near-work on digital displays induces visual fatigue, medically recognized as Digital Eye Strain (DES). While long-term optical health risks remain under study, immediate symptoms include eye dryness, blurred vision, and headaches. This micro-paper isolates the temporal factor of device usage, testing the immediate impact of scheduled micro-breaks on acute symptom prevention.

2. Methodology

A randomized cohort of healthy adults (N=40, ages 20–30) completed an 8-hour document editing task on identical 6.1-inch OLED mobile displays (brightness locked at 300 nits, 40cm viewing distance). Group A (n=20) received automated haptic prompts to look at a target 20 feet away for 20 seconds every 20 minutes. Group B (n=20) worked continuously without enforced breaks. Visual fatigue was measured at baseline and hour 8 using a standardized 10-point Visual Analog Scale (VAS).

3. Results & Data

Mean VAS fatigue scores escalated in both cohorts over the 8-hour testing trial. However, the intervention group demonstrated a significantly slower rate of fatigue accumulation. Group A finished with a mean fatigue score of 3.8, while Group B reached a mean score of 5.8.

Cohort Group	Baseline VAS Score (±SD)	Final 8-Hr VAS Score (±SD)	Net Change (Δ)
Group A (Micro-breaks)	1.2 (±0.3)	3.8 (±0.6)	+2.6
Group B (Control)	1.1 (±0.4)	5.8 (±0.9)	+4.7

4. Discussion & Conclusion

The results confirm that periodic ciliary muscle relaxation significantly blunts the onset of digital eye strain. Enforcing brief 20-second cognitive and visual pivots offsets continuous accommodative demand. Future micro-studies should explore whether display size alterations or specialized blue-light filtration layers further optimize these baseline findings.

References

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- [2] Wong, K. L. (2024). Efficacy of the 20-20-20 rule: A systematic review. *Ophthalmic Review Letters*, 8(1), 45-49.